A proof of the conjectured column recurrences for OEIS A246479, formalized in Lean 4

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Abstract

Let $T(n,k)$ be the number of arrays of length $n+3$ over $\{0,\dots,k\}$ in which no two entries among any four consecutive terms sum to exactly k (OEIS A246479 [1]). R. H. Hardin recorded column recurrences for $k \le 9$ as "Empirical", and Robert Israel (2024) conjectured two recurrences valid for all k: an order-4 recurrence for odd $k \ge 5$ and an order-10 recurrence for even $k \ge 6$. We prove both conjectures, in fact for every odd and every even k, together with Hardin's column formulas for $k = 1,\dots,9$ and Colin Barker's conjectured closed forms for rows 1, 2 and 3 (A246480–A246482). The proof is a transfer-matrix argument on eleven "types" of triples, uniform in k, and is machine-checked in Lean 4 with Mathlib [5]: no `sorry`, no `native_decide`, and only the axioms `propext`, `Classical.choice` and `Quot.sound`.

1 Introduction

OEIS A246479 (R. H. Hardin, 2014) tabulates $T(n,k)$, the number of words $w_1 \cdots w_{n+3}$ over $\{0,\dots,k\}$ with $w_i + w_j \ne k$ whenever $1 \le j - i \le 3$. The entry lists, as "Empirical for column k" (with $a(n) = T(n,k)$),

$$k = 1 : a(n) = a(n-1), \qquad k = 2 : a(n) = a(n-1) + a(n-4),$$
$$k = 3 : a(n) = 2a(n-1) + a(n-3),$$
$$k = 4 : a(n) = 2a(n-1) + a(n-3) + 14a(n-4) + 3a(n-5) + 6a(n-6) + a(n-8) + a(n-9),$$
$$k = 5,7,9 : a(n) = (k-2)a(n-1) + 2a(n-2) + (k-2)a(n-3) + a(n-4),$$

and "order 10" for $k = 6, 8$ (the $k = 6$ recurrence is written out in A246477). In November 2024 Robert Israel added to the entry:

> It appears that for $k \ge 5$ odd, $a(n) = (k-2)a(n-1)+2a(n-2)+(k-2)a(n-3)+a(n-4)$, and that for $k \ge 6$ even, $a(n) = (k-3)a(n-1) + 2a(n-2) + (k-3)a(n-3) + (k^3 - 6k^2 + 15k - 13)a(n-4) + (3k^2 - 11k + 13)a(n-5) + (k^3 - 7k^2 + 19k - 19)a(n-6) + (k^2 - 4k + 6)a(n-7) + a(n-8) + (k-2)a(n-9) + a(n-10)$.

We prove both statements for all k of the given parity, and all of Hardin's column formulas.

2 Related work

The table and its column and row recurrences were contributed by Hardin in 2014 [1]; Israel's two conjectures and a Maple program computing columns by a transfer matrix on all valid triples

were added in 2024 [1]. Columns $k = 2, \ldots, 7$ are the entries A246473–A246478, where Colin Barker (2018) added empirical generating functions; rows 1–3 are A246480–A246482, where Barker conjectured quasi-polynomial closed forms [2]. That each column satisfies *some* linear recurrence is immediate from the transfer-matrix method (Stanley [4], §4.7); the content of Israel's conjecture is the explicit, uniform-in-k recurrences, whose order does not grow with k although the naive transfer matrix has $\Theta(k^3)$ states. To our knowledge no proof had been published: the OEIS entries mark all these formulas as empirical or conjectural, and an OEIS and web search on October 12, 2026 found none. The method is a direct extension of the four-type argument we used for the three-term analogue A245995 [3]; what is new here is the eleven-type reduction for windows of four, the resulting proofs of Israel's and Barker's conjectures, and the machine-checked formalization in Lean 4 [6] and Mathlib [5].

3 Statement

Theorem 1. *For all $n \geq 0$ (here $T(0,k)$ counts words of length 3), over the integers:*

1. *if k is odd, $T(n+4,k) = (k-2)T(n+3,k) + 2T(n+2,k) + (k-2)T(n+1,k) + T(n,k)$;*
2. *if k is even,*

$$\begin{aligned} T(n+10,k) &= (k-3)T(n+9,k) + 2T(n+8,k) + (k-3)T(n+7,k) \\ &+ (k^3-6k^2+15k-13)T(n+6,k) + (3k^2-11k+13)T(n+5,k) + (k^3-7k^2+19k-19)T(n+4,k) \\ &+ (k^2-4k+6)T(n+3,k) + T(n+2,k) + (k-2)T(n+1,k) + T(n,k). \end{aligned}$$

Corollary 2. *Israel's two conjectures hold, and all of Hardin's column recurrences for $k = 1, \ldots, 9$ hold (OEIS offset $n \geq 1$, wherever the right side is defined).*

Theorem 3. *For even k: $T(1,k) = k^4 - 2k^3 + 3k^2 - k$, $T(2,k) = k^5 - 4k^4 + 10k^3 - 11k^2 + 5k$, $T(3,k) = k^6 - 6k^5 + 21k^4 - 40k^3 + 43k^2 - 20k$. For odd k: $T(1,k) = k^4 - 2k^3 + 3k^2 + 3k - 3$, $T(2,k) = k^5 - 4k^4 + 10k^3 - 4k^2 - 10k + 9$, $T(3,k) = k^6 - 6k^5 + 21k^4 - 30k^3 + 5k^2 + 37k - 26$. These are Barker's conjectured formulas for A246480, A246481, A246482.*

4 Proof

Write $\bar{a} = k - a$; a pair at distance at most 3 is forbidden iff its letters are $a, \bar{a}$. Let $\varepsilon = 1$ if k is even and 0 otherwise; for even k, $h = k/2$ is the unique letter with $\bar{h} = h$ (the "middle" letter). Let $N_L(x,y,z)$ be the number of good words of length $L+3$ beginning x, y, z.

Step 1 (prepending a letter). A word $c\,v$ is good iff v is good and $c + v_i \neq k$ for $i = 1, 2, 3$. Hence

$$N_{L+1}(c,x,y) = [\,x \neq \bar{c},\ y \neq \bar{c}\,]\Big(\sum_a N_L(x,y,a) - N_L(x,y,\bar{c})\Big).$$

Step 2 (eleven types). A triple is *valid* if no two of its letters are complementary; in a valid triple the middle letter occurs at most once, and equal letters are non-middle. Classify a valid triple (x,y,z) by its equality pattern and the position of a middle letter: A: $x = y = z$; B, B_m: $x = y \neq z$ with z non-middle, resp. middle; C, C_m: $x = z \neq y$ with y non-middle, resp. middle; D, D_m: $y = z \neq x$ with x non-middle, resp. middle; E: all distinct and non-middle; E_1, E_2, E_3: all distinct with the middle letter in position $1, 2, 3$. We claim $N_L(x,y,z)$ is a function τ_L of the type (0 if invalid), with $\tau_0 \equiv 1$.

Pair sums. Fix a non-forbidden pair (x, y) and sum $\tau_L(x, y, a)$ over a. The letters $a \in \{\bar{x}, \bar{y}\}$ contribute 0; $a = x$, $a = y$ and $a = h$ contribute single type values; all other letters give the same type. This yields, with $\kappa = k - 1 - \varepsilon$,

$$\begin{aligned} S_{xx} &= A + \varepsilon B_m + \kappa B && (x = y), \\ S_{\neq} &= C + D + \varepsilon E_3 + (\kappa - 2)E && (x \neq y, \text{ both non-middle}), \\ S_{h\cdot} &= D_m + \kappa E_1 && (x = h), \qquad S_{\cdot h} = C_m + \kappa E_2 \quad (y = h). \end{aligned}$$

Correction term. By Step 1, $N_{L+1}(c, x, y)$ is the pair sum of (x, y) minus $N_L(x, y, \bar{c})$, and the type of $(x, y, \bar{c})$ is determined by the type of (c, x, y). Checking the eleven cases gives the linear step

$$\begin{aligned} A' &= S_{xx}, & B' &= C' = S_{\neq}, & B'_m &= S_{\cdot h}, & C'_m &= S_{h\cdot}, \\ D' &= S_{xx} - B, & D'_m &= S_{xx} - B_m, & E' &= S_{\neq} - E, & E'_1 &= S_{\neq} - E_3, \\ E'_2 &= S_{h\cdot} - E_1, & E'_3 &= S_{\cdot h} - E_2. \end{aligned}$$

For example, if (c, x, y) has type E_2 $(x = h)$ then $\bar{c}$ is a non-middle letter different from $x, y, \bar{x}, \bar{y}$, so $(x, y, \bar{c})$ has type E_1 and $N_{L+1}(c, x, y) = S_{h\cdot} - E_1$.

Step 3 (total). Summing over y and then x by the same counting,

$$T(L, k) = \varepsilon(k + 1 - \varepsilon)S_{h\cdot} + (k + 1 - \varepsilon)(S_{xx} + \varepsilon S_{\cdot h} + \kappa S_{\neq}).$$

Step 4 (recurrences). The type vector evolves by a fixed linear map M whose entries are polynomials in k, and $T(L, k) = F(M^L \mathbf{1})$ for a fixed linear functional F. Theorem 1 is the statement that $F \circ P(M) = 0$ for the corresponding characteristic polynomial P. For odd k $(\varepsilon = 0)$ only A, B, C, D, E enter, and the rows for B and C coincide, which is why the order drops to 4. We write $F \circ M^i$ as explicit linear forms G_i with polynomial coefficients, check $G_i \circ M = G_{i+1}$ for each i, and then check $G_4 = (k - 2)G_3 + 2G_2 + (k - 2)G_1 + G_0$ (odd) and the analogous identity for G_{10} (even). All of these are polynomial identities, verified in Lean by `ring`. Hardin's formulas for $k = 1, \ldots, 4$ are checked the same way with k fixed; those for $k = 5, \ldots, 9$ are specialisations of Theorem 1. Theorem 3 is the evaluation of G_1, G_2, G_3 at $\mathbf{1}$.

Numerical check. The recursion reproduces all 9999 terms of Hardin's b-file for A246479 (antidiagonals up to $k = 140$).

Remark 4. Theorem 1 holds for every k of the given parity, including $k = 1, 3$ and $k = 2, 4$, as an identity over $\mathbb{Z}$. For $k = 6$ the coefficient $k^3 - 7k^2 + 19k - 19 = 11$ is positive, but read in natural-number arithmetic with truncated subtraction the expression would be wrong, which is why we state the even case over $\mathbb{Z}$.

5 Formalization

All declarations are in namespace `A246479`; see Table 1.

6 Reproducibility

- Source code: `https://github.com/het4rk/OEIS-A246479-proof`. The Lean project is in `A246479/`.
- Toolchain: `leanprover/lean4:v4.35.0-rc4`. Mathlib: tag `v4.35.0-rc4`, commit `1f414401 f69059aa7eead47b53ee40bd38455eeb`, pinned in `lake-manifest.json`.

Lean name (file)	Statement
`Good`, `T`, `N` (`Basic.lean`)	OEIS condition (natural-number sums); $T(n,k)$; $N_L(x,y,z)$
`good_cons`, `N_succ`	Step 1
`typ`, `sum_spec`, `pair_sum` (`Types.lean`)	types and pair sums
`step`, `typ_step`, `N_eq_typ`, `T_eq_F` (`Step.lean`)	Steps 2–3
`Go0`–`Go4`, `Ge0`–`Ge10`, chain lemmas (`Forms.lean`)	Step 4
`T_odd_int`, `T_even_int`, `T_odd` (`Main.lean`)	Theorem 1
`israel_odd`, `israel_even`, `T_col1`–`T_col9`	Corollary 2, verbatim OEIS form
`row1_even`, ..., `row3_odd` (`Rows.lean`)	Theorem 3
`Values.lean`	kernel-checked values $T(1,1)=2$, $T(1,2)=10$, $T(2,2)=14$, $T(1,3)=60$

Table 1: Theorems and their Lean names.

- Build:

```
cd A246479
lake exe cache get
lake build
lake env lean Axioms.lean
```

- Axioms: all 21 theorems listed in `Axioms.lean` depend only on `propext`, `Classical.choice` and `Quot.sound`.
- Attestation: `verification/` contains the build log, the axiom output and `VERIFICATION.md` with SHA-256 hashes of the Lean sources. Continuous integration rebuilds on every push, rejects `sorry` and non-standard axioms, and uploads the logs as artifacts. `scripts/check_bfile.py` re-checks the b-file.

7 Limitations

The argument is a transfer-matrix computation; its interest is that it settles Israel's uniform-in-k conjectures and a family of empirical OEIS formulas with a machine-checked proof. We do not prove that the recurrences have minimal order for every k. We do not formalize Hardin's row recurrences for $n \geq 4$ or Barker's column generating functions, although the latter follow from the recurrences and the initial terms, and each row is a quasi-polynomial in k by the same method. Windows of length five (A246737) and longer are not treated.

References

[1] R. H. Hardin, R. Israel, *A246479: $T(n,k)$ = number of length $n+3$ $0..k$ arrays with no pair in any consecutive four terms totalling exactly k*, The On-Line Encyclopedia of Integer Sequences, 2014 (comments 2024), `https://oeis.org/A246479`.

[2] R. H. Hardin, C. Barker, *A246473–A246478* (columns $k = 2, \ldots, 7$) and *A246480–A246482* (rows 1–3) of A246479, The On-Line Encyclopedia of Integer Sequences, `https://oeis.org/A246480`.

[3] H. Shah (AI-assisted by Grok), *A proof of Hardin's empirical column recurrences for OEIS A245995, formalized in Lean 4*, 2026, `https://github.com/het4rk/OEIS-A245995-proof`.

[4] R. P. Stanley, *Enumerative Combinatorics, Vol. 1*, 2nd ed., Cambridge University Press, 2012.

[5] The mathlib Community, *The Lean mathematical library*, in Proc. CPP 2020, ACM, 367–381.

[6] L. de Moura, S. Ullrich, *The Lean 4 theorem prover and programming language*, in CADE-28, LNCS 12699, Springer, 2021, 625–635.